DBMS ASSIGNMENT

ASSIGNMENT 5

Database Creation and Table Implementation

Project: Library Loan Management System

Database: MySQL 8.0

Objective

Create the Library Loan Management System database and implement all required tables using SQL DDL.

Note: outputs are expected results based on the included sample data. Run the SQL in your MySQL environment and insert your own screenshots if required by your instructor.

1. Objective and Environment

The implementation targets MySQL 8.0. The schema includes six tables: Members, Books, Authors, BookAuthors, BookCopies, and Loans. The generated OpenCopyID column and unique constraint prevent more than one open loan for a physical copy.

2. Entity Relationship Summary

Members 1—many Loans; BookCopies 1—many historical Loans; Books 1—many BookCopies; Books many—many Authors through BookAuthors. A book edition may have several physical copies. Each loan records one member borrowing one physical copy.

3. SQL DDL

```
CREATE DATABASE IF NOT EXISTS LibraryLoanDB;
USE LibraryLoanDB;

CREATE TABLE Members (
    MemberID INT PRIMARY KEY,
    FullName VARCHAR(100) NOT NULL,
    Email VARCHAR(120) NOT NULL UNIQUE,
    Phone VARCHAR(20) UNIQUE,
    JoinedOn DATE NOT NULL,
    MembershipStatus VARCHAR(10) NOT NULL DEFAULT 'Active',
    CONSTRAINT chk_membership_status
        CHECK (MembershipStatus IN ('Active', 'Inactive'))
);

CREATE TABLE Books (
    BookID INT PRIMARY KEY,
    ISBN VARCHAR(20) UNIQUE,
    Title VARCHAR(200) NOT NULL,
    Edition VARCHAR(30),
    PublicationYear INT,
    Category VARCHAR(60) NOT NULL,
    CONSTRAINT chk_publication_year
        CHECK (PublicationYear IS NULL OR PublicationYear BETWEEN 1450 AND 2100)
);

CREATE TABLE Authors (
    AuthorID INT PRIMARY KEY,
    AuthorName VARCHAR(100) NOT NULL
);

CREATE TABLE BookAuthors (
    BookID INT NOT NULL,
    AuthorID INT NOT NULL,
    PRIMARY KEY (BookID, AuthorID),
    CONSTRAINT fk_bookauthors_book FOREIGN KEY (BookID)
        REFERENCES Books(BookID),
    CONSTRAINT fk_bookauthors_author FOREIGN KEY (AuthorID)
        REFERENCES Authors(AuthorID)
);

CREATE TABLE BookCopies (
    CopyID INT PRIMARY KEY,
    BookID INT NOT NULL,
    AccessionNumber VARCHAR(30) NOT NULL UNIQUE,
    ShelfLocation VARCHAR(30) NOT NULL,
    ConditionStatus VARCHAR(12) NOT NULL DEFAULT 'Good',
    CONSTRAINT chk_copy_condition
        CHECK (ConditionStatus IN ('Good', 'Damaged', 'Lost', 'Withdrawn')),
    CONSTRAINT fk_copy_book FOREIGN KEY (BookID)
        REFERENCES Books(BookID)
);

CREATE TABLE Loans (
    LoanID INT PRIMARY KEY,
    MemberID INT NOT NULL,
    CopyID INT NOT NULL,
    BorrowedOn DATE NOT NULL,
    DueOn DATE NOT NULL,
    ReturnedOn DATE NULL,
    OpenCopyID INT GENERATED ALWAYS AS
        (CASE WHEN ReturnedOn IS NULL THEN CopyID ELSE NULL END) STORED,
    CONSTRAINT chk_loan_dates CHECK (DueOn >= BorrowedOn),
    CONSTRAINT chk_return_date CHECK (ReturnedOn IS NULL OR ReturnedOn >= BorrowedOn),
    CONSTRAINT fk_loan_member FOREIGN KEY (MemberID)
        REFERENCES Members(MemberID),
    CONSTRAINT fk_loan_copy FOREIGN KEY (CopyID)
        REFERENCES BookCopies(CopyID),
    CONSTRAINT uq_one_open_loan_per_copy UNIQUE (OpenCopyID)
);
```

4. Table Purpose

Members stores membership and contact details. Books stores bibliographic details for each edition. Authors stores author identities. BookAuthors maps books to authors. BookCopies tracks each physical copy, accession number, shelf, and condition. Loans records borrowing and return history.

5. Verification Commands

```
USE LibraryLoanDB;
SHOW TABLES;
DESCRIBE Members;
DESCRIBE Books;
DESCRIBE Authors;
DESCRIBE BookAuthors;
DESCRIBE BookCopies;
DESCRIBE Loans;
SHOW CREATE TABLE Loans;

Expected result: six tables appear. SHOW CREATE TABLE Loans should show the foreign keys, date checks,
generated OpenCopyID, and unique constraint. The generated column syntax requires a compatible MySQL 8.0
version.
```

6. Implementation Order

Create the database, then Members, Books, Authors, BookAuthors, BookCopies, and Loans. Parent tables must exist before foreign keys reference them. Run the DDL in a fresh schema or back up data before making structural changes.

7. Conclusion

The DDL establishes the core entities, many-to-many author relationship, physical-copy inventory, and loan history needed for the library application.